Lekhraj Mahajan

Ahmedabad, Gujarat

+91 6354245706 | lekhrajmahajan506@gmail.com | linkedin.com/in/lekhrajmahajan | github.com/LekhrajMahajan | **Portfolio Link**

Education

Silver Oak University , Ahemdabad 2023 - 2026
Bachelor of Technology – Computer Engineering , CGPA - 8.60

Government Polytechnic , Bhuj 2020 - 2023
Diploma – Computer Engineering , CGPA - 7.75

Work Experience

MERN Stack Developer - Tech Creature Solutions May 2026 – Present
Full-stack work on The Spot Exam, an enterprise examination SaaS - Ongoing Project Gandhinagar, Gujarat

- Developed the **Node.js / Express / TypeScript** backend with **80+ modular domain modules** (controller–service–repository pattern) on **MongoDB/Mongoose**, with **Zod** request validation and centralized error handling.
- Implemented **JWT authentication** and a granular **RBAC / permission system across 10+ roles** (Master Admin, Company Admin, Exam Manager, Center Manager, Observer, Paper Setter, Biometric Verifier, Candidate), with **audit logging, rate limiting** and Helmet-based security hardening.
- Built **real-time proctoring and live monitoring** with **Socket.IO** (Redis adapter for multi-instance scaling); used **BullMQ** job queues, **node-cron** schedulers and **Redis caching** for background processing and performance.
- Built the **React + TypeScript** admin portal with role-specific layouts, **Zustand** state, **React Hook Form + Zod**, route-level **code splitting (lazy loading)** and **PWA** support.
- Developed the candidate exam client (**React, Vite**) with in-browser **face verification** (**face-api.js**) and violation detection – tab switch, focus loss, copy/paste, no-face and multiple-face events.

Projects

The Spot Exam | *Node.js, Express, TypeScript, MongoDB, Redis, BullMQ, Socket.IO, React, Zustand, Tailwind CSS*
Multi-tenant online examination, proctoring and exam-center management platform Live: thespotexam

- Designed a **multi-tenant SaaS** architecture for coaching companies and government-style exam bodies: three apps (admin portal, candidate exam client, REST API) with **subscription plans, billing and invoicing**.
- Built the exam lifecycle end to end: **question bank, paper setting, shifts, seat allocation with QR codes, bulk candidate import (Excel/CSV)**, live answer capture and secure submission.
- Added integrity layer: **face verification, per-candidate trust score, geo-monitoring of exam centers**, live candidate tracking and violation logs for observers.
- Integrated **Razorpay** payments, **Nodemailer**, Firebase push notifications and **Cloudinary**; added support tickets, **API metrics**, **Winston logging** and **PM2 deployment**.

AURA Headphones | *React, Node.js, Express, MongoDB, Redis, Firebase Auth, GSAP, Cloudinary*
Full-stack premium audio e-commerce platform with AI product recommendations Live: auraheadphones

- Built a full-stack **e-commerce** flow (catalog, product details, cart, checkout, order tracking, user profile) with a React frontend on **Vercel** and **Node/Express REST API** on Render.
- Implemented **Firebase Authentication** (email/password and Google sign-in) with server-side **ID-token verification**, **JWT**-protected routes and a separate admin role middleware.
- Improved performance with **Redis caching** (1-hour TTL) for product APIs, **express-rate-limit**, Cloudinary image optimization and **lazy-loaded routes**; added **GSAP/Lenis** scroll animations.
- Built an **admin dashboard** to manage products, orders, users and **discount coupons**.

Technical Skills

Languages : TypeScript, JavaScript (ES6+), HTML5, CSS3
Frontend : React.js, Vite, Tailwind CSS, Zustand, React Hook Form, GSAP, PWA
Backend : Node.js, Express.js, REST API Design, MVC / Modular Architecture, Socket.IO, WebSocket, BullMQ, node-cron, Zod
Databases : MongoDB, Mongoose, Redis (caching)
Auth & Security : JWT, bcrypt, RBAC, Rate Limiting, Helmet, Audit Logging, Face Verification (face-api.js)
Integrations : Razorpay, Nodemailer, Firebase (Auth, FCM), Cloudinary
Tools & Deployment : Git, GitHub, Postman, VS Code, PM2, Vercel, Render, Netlify